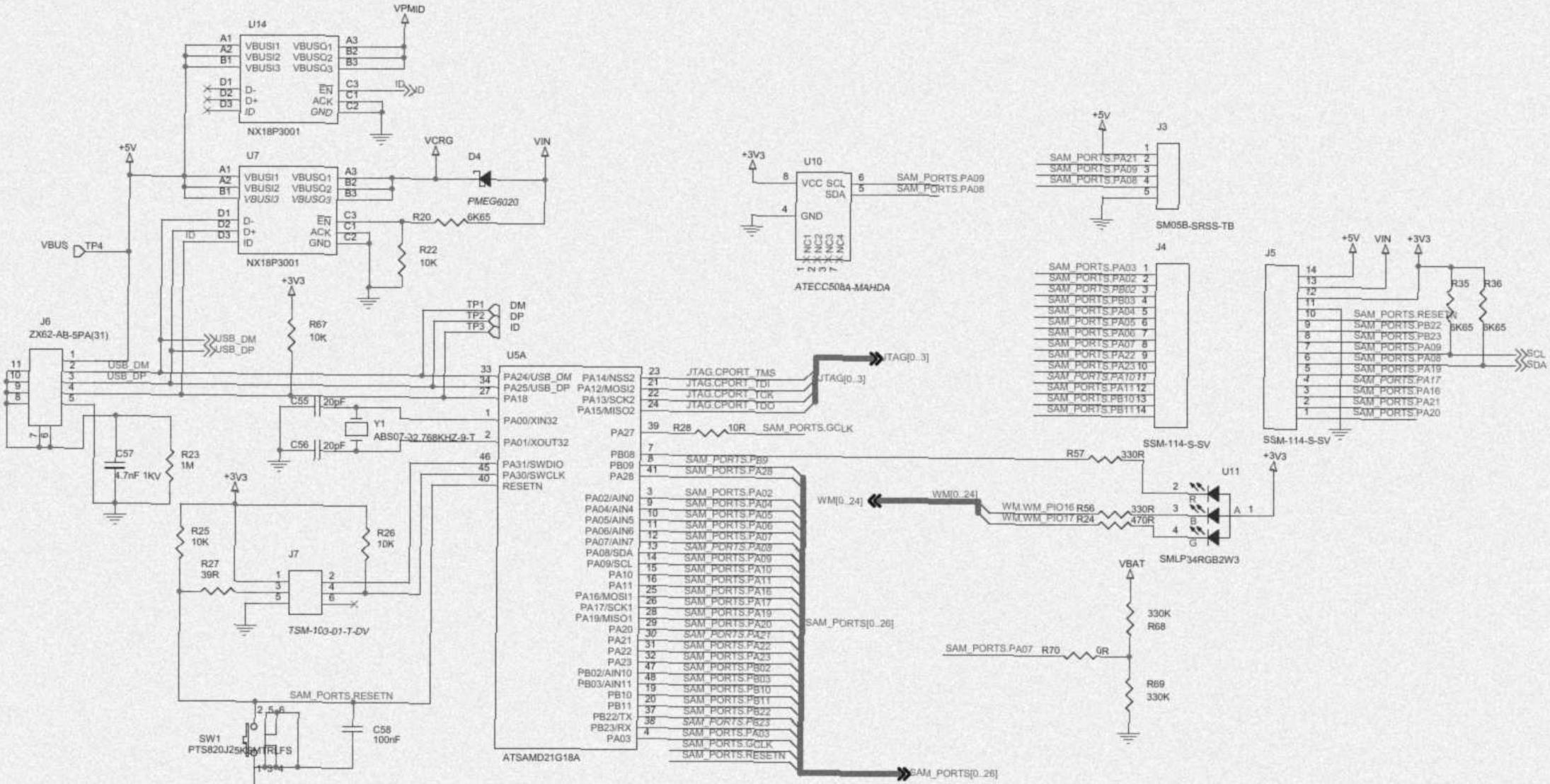


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sam_ports[0..26]

SAM_PORTS[0..26]

SAM_PORTS.PA02
SAM_PORTS.PA03
SAM_PORTS.PA04
SAM_PORTS.PA05
SAM_PORTS.PA06
SAM_PORTS.PA07
SAM_PORTS.PA08
SAM_PORTS.PA09
SAM_PORTS.PA10
SAM_PORTS.PA11
SAM_PORTS.PA16
SAM_PORTS.PA17
SAM_PORTS.PA19
SAM_PORTS.PA20
SAM_PORTS.PA21
SAM_PORTS.PA22
SAM_PORTS.PA23
SAM_PORTS.PB02
SAM_PORTS.PB03
SAM_PORTS.PB10
SAM_PORTS.PB11
SAM_PORTS.PB22
SAM_PORTS.PB23
SAM_PORTS.GCLK
SAM_PORTS.RESETN
SAM_PORTS.PB9
SAM_PORTS.PA28

MIPI_D0N R38 100R MIPI_D0P
MIPI_D1N R39 100R MIPI_D1P
MIPI_CLKN R40 100R MIPI_CLKP

J8
+3V3
MIPI_D0N
MIPI_D0P
MIPI_D1N
MIPI_D1P
MIPI_CLKP
MIPI_CLKN
MIPI_GP0
MIPI_GPT
MIPI_SCL
MIPI_SDA
R72 1K8
R71 1K8
SFW15R-1STE1LF

U1-2
SAM_PORTS.PA03 B1
SAM_PORTS.PA02 C2
SAM_PORTS.PA04 D1
SAM_PORTS.PA07 G2
SAM_PORTS.PA22 G1
IO_B1_B1
IO_B1_C2
IO_B1_D1
IO_B1_G2
IO_B1_G1

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U1-3
MIPI_D1P J2
MIPI_D1N J1
K2
K1
MIPI_D0P L2
MIPI_D0N L1
SAM_PORTS.PB9 N2
WM.WM_PIO36 N1
PANEL_SCL K5
PANEL_SDA L4
WM.WM_RESET R1
MIPI_SDA P2
MIPI_SCL P1
IO_B2_J2
IO_B2_J1
IO_B2_K2
IO_B2_K1
IO_B2_L2
IO_B2_L1
IO_B2_N2
IO_B2_N1
IO_B2_K5/RUP1
IO_B2_L4/RDN1
IO_B2_R1
IO_B2_P2
IO_B2_P1

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U1-4
SAM_PORTS.PA23 N3
SAM_PORTS.PA10 P3
SAM_PORTS.PA11 R3
SAM_PORTS.PB10 T3
SAM_PORTS.PB11 T2
WM.WM_PIO35 R4
WM.WM_PIO31 T4
WM.WM_PIO8 N5
WM.WM_PIO5 N6
PEX_PIN7 M6
MIPI_GP0 M7
WM.WM_PIO20 R5
WM.WM_PIO18 T5
WM.WM_PIO21 R6
WM.WM_RX T6
PEX_PIN8 L7
CFG_HOLD R7
CFG_WP T7
PEX_PIN14 L8
PEX_PIN12 M8
PEX_PIN10 N8
PEX_PIN6 P8
IO_B3_N3
IO_B3_P3
IO_B3_R3
IO_B3_T3
IO_B3_T2
IO_B3_R4/PLL1_CLKOUTP
IO_B3_T4/PLL1_CLKOUTN
IO_B3_N5
IO_B3_N6
IO_B3_M6
IO_B3_M7
IO_B3_R5
IO_B3_T5
IO_B3_R6
IO_B3_T6
IO_B3_L7
IO_B3_R7
IO_B3_T7
IO_B3_L8
IO_B3_M8
IO_B3_N8
IO_B3_P8

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PANEL_HPD R76 6K65
R77 6K65
PANEL_TX2P C69 100nF
PANEL_TX2N C70 100nF
PANEL_TX1P C71 100nF
PANEL_TX1N C72 100nF
PANEL_TX0P C73 100nF
PANEL_TX0N C74 100nF
PANEL_CLKP C75 100nF
PANEL_CLKN C76 100nF
+3V3
R75 6K65
R74 6K65
+5V
PANEL_SCL
PANEL_SDA
J13
10118241-001RLF

+3V3
U3
CFG_DI
CFG_DCLK
CFG_HOLD
CFG_WP
CFG_CE
SI
SCLK
HOLD
WP
CS
VCC
GND
SO
R4 10R CFG_DO
W25Q16DVSNIG
R79 6K65
R78 6K65

WM.WM_PIO27 N9
PEX_PIN1 R10
WM.WM_PIO29 T10
PEX_PIN5 R11
WM.WM_PIO1 T11
PEX_PIN30 R12
PEX_RST T12
MIPI_GP1 P9
PEX_PIN42 R13
PEX_PIN28 T13
PEX_PIN16 M10
WM.WM_PIO28 N11
PEX_PIN47 T14
PEX_PIN45 T15
PEX_PIN20 N12
PEX_PIN44 P14
PEX_PIN46 R14

U1-5
IO_B4_N9
IO_B4_R10
IO_B4_T10
IO_B4_R11
IO_B4_T11
IO_B4_R12
IO_B4_T12
IO_B4_P9
IO_B4_R13
IO_B4_T13
IO_B4_M10/RUP2
IO_B4_N11/RDN2
IO_B4_T14
IO_B4_T15
IO_B4_N12
IO_B4_P14/PLL4_CLKOUTP
IO_B4_R14/PLL4_CLKOUTN

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U1-6
N14
P15
P16
PANEL_TX2N R16
PANEL_TX2P R16
PANEL_CLKN N16
PANEL_CLKP N15
L13
SAM_PORTS.PA28 L16
L15
PANEL_TX1N K16
PANEL_TX1P K15
J14
WM.WM_PIO32 J13
IO_B5_N14/RUP3
IO_B5_P15/RDN3
IO_B5_P16
IO_B5_R16
IO_B5_N16
IO_B5_N15
IO_B5_L13
IO_B5_L16
IO_B5_L15
IO_B5_K16
IO_B5_K15
IO_B5_J14
IO_B5_J13

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U1-7
PEX_PIN32 F13
SAM_PORTS.PA09 B16
PEX_PIN49 D16
PEX_PIN51 D15
SAM_PORTS.PA19 C16
SAM_PORTS.PA08 C15
IO_B6_F13
IO_B6_B16
IO_B6_D16
IO_B6_D15
IO_B6_C16
IO_B6_C15

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SDRAM_A5 C14
SDRAM_A4 D14
SDRAM_A6 D11
SDRAM_A7 D12
SAM_PORTS.PB22 A13
SDRAM_A10 B13
SDRAM_A3 A14
SDRAM_CK B14
SDRAM_A8 E11
SDRAM_A11 E10
SDRAM_A0 A12
SDRAM_A1 B12
SDRAM_CS A11
SDRAM_WE B11
SDRAM_A2 A15
SDRAM_UDQM F9
SDRAM_BS0 A10
SDRAM_BS1 B10
SDRAM_A9 C9
SDRAM_RAS D9
SDRAM_CKE E9

U1-8
IO_B7_C14
IO_B7_D14
IO_B7_D11
IO_B7_D12
IO_B7_A13
IO_B7_B13
IO_B7_A14/PLL2_CLKOUTN
IO_B7_B14/PLL2_CLKOUTP
IO_B7_E11/RUP4
IO_B7_E10/RDN4
IO_B7_A12
IO_B7_B12
IO_B7_A11
IO_B7_B11
IO_B7_A15
IO_B7_F9
IO_B7_A10
IO_B7_B10
IO_B7_C9
IO_B7_D9
IO_B7_E9

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SDRAM_D9 C8
SDRAM_D13 D8
SDRAM_LDQM A7
SDRAM_D7 A6
SDRAM_D15 B6
SDRAM_D6 B5
SDRAM_D14 D6
SDRAM_D4 A4
SDRAM_D1 B4
SDRAM_D0 A2
SDRAM_D3 A3
SDRAM_D2 B3
SAM_PORTS.PB02 C3
SAM_PORTS.PA05 D3

U1-9
IO_B8_C8
IO_B8_D8
IO_B8_A7
IO_B8_A6
IO_B8_B6
IO_B8_B5
IO_B8_D6
IO_B8_A4
IO_B8_B4
IO_B8_A2
IO_B8_A3
IO_B8_B3
IO_B8_C3/PLL3_CLKOUTN
IO_B8_D3/PLL3_CLKOUTP

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WM[0..24]

J11
+3V3
JTAG.CPORT_TMS
JTAG.CPORT_TCK
JTAG.CPORT_TDO
JTAG.CPORT_TDI
FTSH-105-01-F-D-K

J2
WM.WM_PIO2 PEX_PIN1
WM.WM_PIO3 PEX_PIN3
WM.WM_PIO4 PEX_PIN5
WM.WM_PIO34 PEX_PIN7
PEX_PIN11
PEX_PIN13
WM.WM_PIO24
WM.WM_PIO25
PEX_PIN23
PEX_PIN25
PEX_PIN31
PEX_PIN33
PEX_PIN45
PEX_PIN47
PEX_PIN49
PEX_PIN51
PIN1 PIN2 PIN3 PIN4 PIN5 PIN6 PIN7 PIN8 PIN9 PIN10 PIN11 PIN12 PIN13 PIN14 PIN15 PIN16 PIN17 PIN18 PIN19 PIN20 PIN21 PIN22 PIN23 PIN24 PIN25 PIN26 PIN27 PIN28 PIN29 PIN30 PIN31 PIN32 PIN33 PIN34 PIN35 PIN36 PIN37 PIN38 PIN39 PIN40 PIN41 PIN42 PIN43 PIN44 PIN45 PIN46 PIN47 PIN48 PIN49 PIN50 PIN51 PIN52
MINI PCIe
PEX_PIN6
PEX_PIN8
PEX_PIN10
PEX_PIN12
PEX_PIN14
PEX_PIN16
PEX_PIN20
PEX_RST
PEX_PIN26
PEX_PIN30
PEX_PIN32
PEX_PIN42
PEX_PIN44
PEX_PIN46
PEX_PIN48
R43 6K65
R41 6K65
R42 6K65
R5 0R
R6 0R
USB_DM
USB_DP
place close to USB connector

WM.WM_PIO1
WM.WM_PIO2
WM.WM_PIO3
WM.WM_PIO4
WM.WM_PIO5
WM.WM_PIO7
WM.WM_PIO8
WM.WM_PIO16
WM.WM_PIO17
WM.WM_PIO18
WM.WM_PIO20
WM.WM_PIO21
WM.WM_TX
WM.WM_RX
WM.WM_PIO24
WM.WM_PIO25
WM.WM_PIO27
WM.WM_PIO28
WM.WM_PIO29
WM.WM_PIO31
WM.WM_PIO32
WM.WM_PIO34
WM.WM_PIO35
WM.WM_PIO36
WM.WM_RESET

U1-10
SAM_PORTS.GCLK E2
SAM_PORTS.RESETN E1
MIPI_CLKP M2
MIPI_CLKN M1
PEX_PIN13 R8
PEX_PIN11 T8
PEX_PIN25 R9
PEX_PIN23 T9
PANEL_HPD M16
M15
E16
WM.WM_TX E15
PEX_PIN31 A9
PEX_PIN33 B9
A8
B8
SAM_PORTS.PA06 F3
L3
WM.WM_PIO7 P6
PEX_PIN3 P11
L14
PEX_PIN48 F14
SAM_PORTS.PB23 C11
SAM_PORTS.PB03 C6
CLK0_B1_E2/DIFFCLK_0P
CLK1_B1_E1/DIFFCLK_0N
CLK2_B2_M2/DIFFCLK_1P
CLK3_B2_M1/DIFFCLK_1N
CLK15_B3_R8/DIFFCLK_6P
CLK14_B3_T8/DIFFCLK_6N
CLK13_B4_R9/DIFFCLK_7P
CLK12_B4_T9/DIFFCLK_7N
CLK7_B5_M16/DIFFCLK_3N
CLK6_B5_M15/DIFFCLK_3P
CLK5_B6_E16/DIFFCLK_2N
CLK4_B6_E15/DIFFCLK_2P
CLK8_B7_A9/DIFFCLK_5N
CLK9_B7_B9/DIFFCLK_5P
CLK10_B8_A8/DIFFCLK_4N
CLK11_B8_B8/DIFFCLK_4P
IO_B1_F3/VREFB1N0
IO_B2_L3/VREFB2N0
IO_B3_P6/VREFB3N0
IO_B4_P11/VREFB4N0
IO_B5_L14/VREFB5N0
IO_B6_F14/VREFB6N0
IO_B7_C11/VREFB7N0
IO_B8_C6/VREFB8N0

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+3V3
R1 10K
R2 10K
R3 10K
JTAG[0..3]
CFG_DI C1
CFG_CE D2
F4
CFG_DCLK H1
CFG_D0 H2
H5
JTAG.CPORT_TDI H4
JTAG.CPORT_TCK H3
JTAG.CPORT_TMS J5
JTAG.CPORT_TDO J4
J3
PANEL_TX0N J16
PANEL_TX0P J15
H14
H13
H12
G12
SAM_PORTS.PA20 G16
SAM_PORTS.PA21 G15
SAM_PORTS.PA16 F16
SAM_PORTS.PA17 F15
SDRAM_D11 E8
SDRAM_D8 F8
SDRAM_CAS B7
SDRAM_D10 E7
SDRAM_D12 E6
SDRAM_D5 A5

U1-11
IO_B1_C1/DATA1,ASDO
IO_B1_D2/FLASH_NCE,NCSO
NSTATUS_B1_F4/NSTATUS
IO_B1_H1/DCLK
IO_B1_H2/DATA0
NCONFIG_B1_H5/NCONFIG
TDI_B1_H4/TDI
TCK_B1_H3/TCK
TMS_B1_J5/TMS
TDO_B1_J4/TDO
NCE_B1_J3/NCE
IO_B5_J16/DEV_OE
IO_B5_J15/DEV_CLRN
CONF_DONE_B6_H14/CONF_DONE
MSEL0_B6_H13/MSEL0
MSEL1_B6_H12/MSEL1
MSEL2_B6_G12/MSEL2
IO_B6_G16/INIT_DONE
IO_B6_G15/CRC_ERROR
IO_B6_F16/NCEO
IO_B6_F15/CLKUSR
IO_B8_E8/DATA2
IO_B8_F8/DATA3
IO_B8_B7/DATA4
IO_B8_E7/DATA5
IO_B8_E6/DATA6
IO_B8_A5/DATA7

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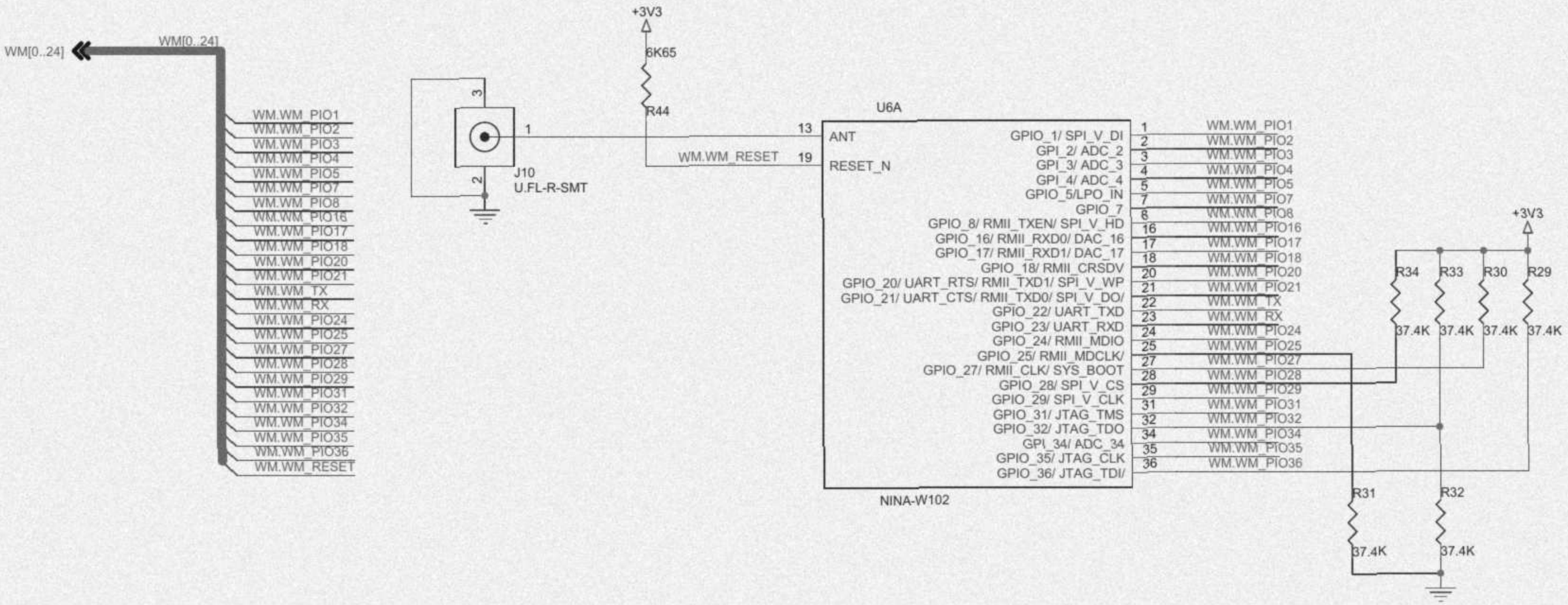
+3V3
U9A
AS4C4M16SA-7BCN
SDRAM_A0 H7
SDRAM_A1 H8
SDRAM_A2 J8
SDRAM_A3 J7
SDRAM_A4 J3
SDRAM_A5 J2
SDRAM_A6 H3
SDRAM_A7 H2
SDRAM_A8 H1
SDRAM_A9 G3
SDRAM_A10 H9
SDRAM_A11 G2
SDRAM_BS0 G7
SDRAM_BS1 G8
SDRAM_CS G9
SDRAM_RAS F8
SDRAM_CAS F7
SDRAM_WE F9
SDRAM_UDQM F1
SDRAM_LDQM E8
A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 BS0 BS1 *CS *RAS *CAS *WE UDQM LDQM
VDD VDD_1 VDD_2 VDDQ VDDQ_1 VDDQ_2 VDDQ_3 VDDQ_4
VSS VSS_1 VSS_2 VSSQ VSSQ_1 VSSQ_2 VSSQ_3
DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7 DQ8 DQ9 DQ10 DQ11 DQ12 DQ13 DQ14 DQ15
A8 B9 B8 C9 C8 D9 D8 E9 E1 D2 D1 C2 C1 B2 B1 A2
SDRAM_D0 SDRAM_D1 SDRAM_D2 SDRAM_D3 SDRAM_D4 SDRAM_D5 SDRAM_D6 SDRAM_D7 SDRAM_D8 SDRAM_D9 SDRAM_D10 SDRAM_D11 SDRAM_D12 SDRAM_D13 SDRAM_D14 SDRAM_D15
CKE F3 SDRAM_CKE
CLK F2 R73 10R SDRAM_CK
NC_1 G1
NC E2

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U4
EZ6301QI
VSYS
SAM_PORTS[0..26]
SAM_PORTS.RESETN
R10 10R
C66 47uF 6V3
C19 27nF
C37 27nF
PVIN
AVIN
SWEN
L1EN
L2EN
SWOK
L1OK
L2OK
SS
BTMP
PG
AGND
PGND
SW
VOUT
VFB
L1IN
L1OUT
L1FB
L2IN
L2OUT
L2FB
SWFB
L1FB
L2FB
+3V3
C1 27pF
R7 6K65
R8 169K
R9 37.4K
C2 47uF 6V3
+2V5
TP11 +2V5
C20 27pF
R11 110K
R12 61K9
C21 47uF 6V3
L1FB
+1V2
C22 27pF
R13 110K
R14 332K
C40 47uF 6V3
L2FB

+3V3
TP9 +3V3
C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15 C16 C17 C18 100nF
+1V2
TP10 +1V2
C24 C25 C26 C27 C28 C29 100nF
TP8 GND
+3V3
C30 C31 C32 C33 C34 C35 C36 C38 C39 C59 C60 C61 100nF
VPLL
C41 C42 C43 C44 100nF
VCCA
C45 C46 C47 C48 100nF
VB5
C49 C50 100nF
VB2
C51 C52 100nF

L1 BLM15AX601SN1#
L2 BLM15AX601SN1#
+2V5
+3V3
+1V2
R15 R16 R17 R18 0R
VB5
VB2
VCCA
VPLL
J12
VCCINT_J12 VCCINT_M11 VCCINT_M9 VCCINT_K10 VCCINT_K9 VCCINT_L6 VCCINT_K11 VCCINT_K7 VCCINT_J6 VCCINT_H11 VCCINT_H8 VCCINT_G10 VCCINT_G9 VCCINT_G8 VCCINT_G7 VCCINT_G6 VCCINT_F11 VCCINT_F7
VCCA4_L12 VCCA3_F5 VCCA2_F12 VCCA1_L5
VCCIO8_C7 VCCIO8_C4 VCCIO8_A1 VCCIO7_C13 VCCIO7_C10 VCCIO7_A16 VCCIO6_G14 VCCIO6_E14 VCCIO5_M14 VCCIO5_K14 VCCIO4_T16 VCCIO4_P13 VCCIO3_T1 VCCIO3_P7 VCCIO2_M3 VCCIO2_K3 VCCIO1_G3 VCCIO1_E3
VCCD_PLL4_N13 VCCD_PLL3_D4 VCCD_PLL2_D13 VCCD_PLL1_N4
U1-1
U1-12
U1-13
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VCRG
R21 470R
D1 HSMG-C190 GREEN
VPMID
C62 47uF 6V3
+3V3
R58 10K
C68 47uF 6V3
U12
BQ24195LRGET
VBUS1 VBUS2 PMID OTG D+ D- STAT SDA SCL INT CE TH
SW1 SW2 BTST REGN PGND1 PGND2 SYS1 SYS2 BAT1 BAT2 ILIM TS1 TS2
L4 SRP4020TA-2R2M
C64 47nF
C65 4.7uF
C67 47uF 6V3
USB_DP R59 0R
USB_DM R60 0R
R62 10K
R63 330R
R65 6K65 REGN
R64 10K
R61 10K
Q1 2N7002
ID
VBAT
F1 MFFSMF050
D3 PMEG6020
J12 S2B-PH-SM4-TB
C63 47uF 6V3

U5B
ATSAMD21G18A
GND0 GND1 EP GNDANA GND2 VDDIO0 VDDIO1 VDDIN VDDANA VDDCORE
L3 BLM15AX601SN1#
C53 100nF
C54 100nF

U6B
NINA-W102
VCC_IO VCC
RSVD1 RSVD2 RSDV3
GND1 GND2 GND3 GND4 GND5
GND_PAD1 GND_PAD2 GND_PAD3 GND_PAD4 GND_PAD5 GND_PAD6 GND_PAD7 GND_PAD8 GND_PAD9 GND_PAD10 GND_PAD11 GND_PAD12

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